ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q2, Q3, Q4

Question 2

(a) $x_1 = \theta$, $x_2 = \theta^{\cdot}$. $x^{\cdot}_1 = x_2$, $x^{\cdot}_2 = (1/J)(\tau - b\, x_2 - mg\ell \sin x_1)$.

Equilibrium means $x^{\cdot} = 0$, so $x_2 = 0$ and $mg\ell \sin \theta_e = \tau_0$. Sence sine is periodic we can always find a θ_e that satisfies this, so an equilibrum exists for any τ_0.

(b) Jacobian at $\theta_e = 60°$: A = [0, 1; $-(mg\ell/J) \cos \theta_e$, $-b/J$].

$mg\ell/J = 2.4525/0.02 = 122.625$; $\cos(60) = -0.9524$, so $A_{21} = -122.625 \times (-0.9524) = +116.8$. $A_{22} = -b/J = -2.5$.

A = [0, 1; 116.8, −2.5], **B = [0; 50]**.

(c) At steady state $\theta^{\cdot} = 0$ and $\theta^{\cdot\cdot} = 0$.

A PD controller is stable and its job is to drive $(\theta_d - \theta)$ to zero, so $\theta_{ss} = \theta_d = 90°$. The gravity term only affects the transient. Hence the steady-state error is **0°** and the joint does come to rest at θ_d.

(d) New steady-state error = **0**, because the added term cancels the gravity torque exactly.

The pols do not change, because the added term is a feed-forward term that does not depend on the error, so it cannot affect the stability of the loop.

Question 3

(a) (1) Taking the difference of two readings amplifies noise.

(2) The high frequency noise in the encoder signal gets magnified when we subtract successive readings and divide by a small T.

(b) y[k] = 0.0591 (u[k] + u[k−1]) + 0.8818 y[k−1]. DC gain = 1.

(c) $A - LC = [-\ell_1, 1; -\ell_2, -2]$. $\det(sI - A + LC) = (s + \ell_1)(s + 2) + \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 + \ell_2)$.

Desiered: $(s + 10)^2 = s^2 + 20s + 100$. Match: $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$; $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 64$. **$L = [18, 64]^T$**.

(d) Advantage of the observer: it uses the plant model and the applied input u, so its velocity estimate has (almost) no phase lag; the 20 Hz filter on differenced data always delays by about $1/(2\pi f_c) \approx 8$ ms. It can also reconstruct states that cannot be obtained by differencing (e.g. disturbance, when augmented).

Fliter is safer when the model is uncertain or changes (unknown payload, friction, contact, acuator saturation): the observer then gives a biased or poorly-behaved estimat while the model-free filter still follows the measured data.

Question 4

(a) Pols: $-2 \pm 2j$. $N_{ff} = 8$.

(b) With b = 0.75 ($N_{ff} = 8$, $k_1 = 2$): $G_{DC} = bN_{ff}/(c + bk_1) = 0.75 \times 8/(6 + 0.75 \times 2) = 6/7.5 = 0.8$.

θ_{ss} = **0.8**. Tracking error = $(1 - 0.8)/1 \times 100$ = **20%**. (Less than the 25% actuator loss, because the stiffness also dropped.)

(c) $\theta^{\cdot\cdot} + a\theta^{\cdot} + c\theta = b(-k_1\theta - k_2\theta^{\cdot} + K_I\xi)$, $\xi^{\cdot} = r - \theta$. Take r = 0 and differentiate once, using $\xi^{\cdot} = -\theta$: $d^3\theta/dt^3 + (a + bk_2)\theta^{\cdot\cdot} + (c + bk_1)\theta^{\cdot} + bK_I\theta = 0$.

Char. poly: **$s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$**.

Desiered: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

Match: $2 + k_2 = 8 \Rightarrow \mathbf{k_2 = 6}$; $6 + k_1 = 24 \Rightarrow \mathbf{k_1 = 18}$; $K_I = 32$ ($b = 1$).

(d) Because the closed loop is stable, the steady-state error is zero for any b and c.

Failure mode: integrater windup. Remedy: clamp the integrator output (anti-windup).